Province of the
EASTERN CAPE
EDUCATION

NATIONAL SENIOR CERTIFICATE/ *NASIONALE SENIOR SERTIFIKAAT*

GRADE/*GRAAD* 10

NOVEMBER 2019

TECHNICAL MATHEMATICS P1/ *TEGNIESE WISKUNDE V1* MARKING GUIDELINE/*NASIENRIGLYN*

MARKS/*PUNTE*: 100

This marking guideline consists of 9 pages./
Hierdie nasienriglyn bestaan uit 9 bladsye.

QUESTION/*VRAAG 1*

Ques./ Vraag	SOLUTION/*OPLOSSING*	EXPLANATION/ *VERDUIDELIKING*	
1.1	$\sqrt{49} < \sqrt{62} < \sqrt{64}$ $\sqrt{62}$ lies between / *lê tussen* 7 and/*en* 8	✓ method / *metode* ✓ answer / *antwoord*	(2)
1.2	2 | 72 2 | 36 r 0 2 | 18 r 0 2 | 9 r 0 2 | 4 r 1 2 | 2 r 0 2 | 1 r 0 0 r 1 ∴ $72_{10} = 1001000_2$	✓ method / *metode* ✓ answer / *antwoord* No mark awarded for answer if base NOT indicated/Geen punt toegeken vir antwoord as basis NIE aangedui is NIE.	(2)
1.3	1110_2 $\times 111_2$ 1110_2 11100_2 111000_2 1100010_2	✓ method / *metode* ✓ answer / *antwoord*	(2)
1.4	$8{,}72 \times 10^{-4}$	✓ answer / *antwoord*	(1)
1.5	$5(-3)^2 + 3(-3)(2) - 2(2)^2 = 19$	✓ substitution / *vervanging* ✓ answer / *antwoord*	(2)
			[9]

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QUESTION/*VRAAG 2*				
Ques./ Vraag	**SOLUTION/*OPLOSSING***		**EXPLANATION/ *VERDUIDELIKING***	
2.1	2.1.1	$5(7x-5y)^2$ $=5(49x^2-70xy+25y^2)$ $=245x^2-350xy+125y^2$	✓ $49x^2$ ✓ $-70xy$ ✓ $25y^2$ ✓ answer simplified / *antwoord vereenvoudig*	(4)
	2.1.2	$(2a+3)(4a^2-6a+9)$ $=8a^3-12a^2+18a+12a^2-18a+27$ $=8a^3+27$	✓ $8a^3$ ✓ 27	(2)
	2.1.3	$(3+4i)(-2-5i)$ $=-6-15i-8i-20i^2$ $=-6-23i-20(-1)$ $=-6-23i+20$ $=14-23i$	✓ expand / *uitbreiding* ✓ substitution / *vervanging* **-1** ✓ simplification / *vereenvoudiging*	(3)
2.2	2.2.1	$\frac{3^{x+1}.81^x}{9^{x+2}}$ $=\frac{3^{x+1}.(3^4)^x}{(3^2)^{x+2}}$ $=\frac{3^{x+1}.3^{4x}}{3^{2x+4}}$ $=3^{x+1+4x-2x-4}$ $=3^{3x-3}$	✓ definition 3^4 and/en 3^2 ✓ power law / magswet ✓ division law / *delingswet* ✓ answer / *antwoord*	(4)
	2.2.2	$\frac{(9x^2)^4\times 3x^2}{27}$ $=\frac{6561x^8\times 3x^2}{27}$ $=\frac{19683x^{10}}{27}$ $=729x^{10}$	✓ power law / *magswet* ✓ simplification / *vereenvoudiging* ✓ answer / *antwoord*	(3)
				[16]

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QUESTION/*VRAAG 3*			
Ques./ Vraag	**SOLUTION/*OPLOSSING***	**EXPLANATION/ *VERDUIDELIKING***	
3.1	$x^2 + 5x - 6$ $= (x + 6)(x - 1)$	✓ $(x + 6)$ ✓ $(x - 1)$	(2)
3.2	$-4a^3 + 32$ $= -4(a^3 - 8)$ $= -4(a - 2)(a^2 + 2a + 4)$	✓-4 common factor / *gemene faktor* ✓ $(a - 2)$ ✓ $(a^2 + 2a + 4)$	(3)
3.3	$\frac{x^2(x+7)-2x(x+7)+(x+7)}{(x+7)(x-1)^2}$ $=\frac{(x+7)(x^2-2x+1)}{(x+7)(x-1)^2}$ $=\frac{(x-1)(x-1)}{(x-1)(x-1)}$ $=1$	✓ $(x + 7)$ common factor / *gemene faktor* ✓(x^2-2x+1) ✓ factorise/faktoriseer: $(x-1)(x-1)$ ✓ simplification / *vereenvoudiging*	(4)
			[9]

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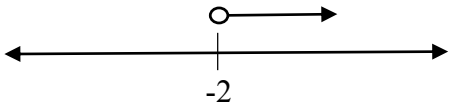
QUESTION/*VRAAG 4*				
Ques./ Vraag	**SOLUTION/*OPLOSSING***		**EXPLANATION/ *VERDUIDELIKING***	
4.1	4.1.1	$9^{x-1} = 81$ $(3^2)^{x-1} = 3^4$ $3^{2x-2} = 3^4$ $\therefore 2x - 2 = 4$ $2x = 6$ $x = 3$	✓ 3^2 and / *en* 3^4 ✓ exponent law / *eksponent wet* ✓ exponents equal / *eksponente gelyk* ✓ answer / *antwoord*	(4)
	4.1.2	$(x + 5)(2x - 3) = 0$ $\therefore x = -5$ or / *of* $x = \frac{3}{2}$	✓ -5 ✓ $\frac{3}{2}$	(2)
	4.1.3	$\frac{2x + 16}{x} = 10$ $2x + 16 = 10x$ $16 = 8x$ $2 = x$	✓ $2x + 16 = 10x$ ✓$16 = 8x$ ✓ answer / *antwoord*	(3)
4.2	$2(x + 4) > x + 6$ $2x + 8 > x + 6$ $x > -2$  		✓ expand / *uitbreiding* ✓ answer / *antwoord* ✓ diagram / *diagram*	(3)
4.3	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{x^2}{a^2} = 1 - \frac{y^2}{b^2}$ $x^2 = \left(1 - \frac{y^2}{b^2}\right) \times a^2$ $x^2 = \frac{b^2 - y^2}{b^2} \times a^2$ $x^2 = \frac{a^2(b^2 - y^2)}{b^2}$ $x = \pm\frac{a}{b}\sqrt{b^2 - y^2}$		✓ $-\frac{y^2}{b^2}$ ✓$\times a^2$ ✓$\frac{a^2(b^2-y^2)}{b^2}$ ✓ answer / *antwoord*	(4)
4.4	4.4.1	$P = 2(l + b)$ $30 = 2(2x + y)$ $P = 2(l + b)$ $24 = 2(x + y)$	✓ answer / *antwoord* ✓ answer / *antwoord*	(2)

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	4.4.2	Substitution/*Vervanging* $30 = 2(2x + y)\ldots$ ❶ $\quad 24 = 2(x + y)\ldots$ ❷ $15 = 2x + y$ $15 - 2x = y\ldots$❸ Sub/*Vervang* ❸in ❷: $24 = 2(x + (15 - 2x))$ $24 = 2(x + 15 - 2x)$ $24 = 2(-x + 15)$ $12 = -x + 15$ $-3 = -x$ $3 = x$ Sub/*Vervang* $x = 3$ in ❸: $15 - 2(3) = y$ $9 = y$ **OR/*OF*** Elimination/*Eliminasie:* $30 = 2(2x + y)\ldots$ ❶ $\quad 24 = 2(x + y)..$ ❷ $30 = 4x + 2y\ldots$❸ $\quad 24 = 2x + 2y\ldots$❹ ❸ – ❹: $30 = 4x + 2y$ $\underline{24 = 2x + 2y}$ $6 = 2x$ $3 = x$ Sub/*Vervang* $x = 3$ in ❸: $30 = 4(3) + 2y$ $30 = 12 + 2y$ $18 = 2y$ $9 = y$	✓ making y the subject / *maak y die onderwerp* ✓ substitution / *vervanging* ✓ expanding / *uitbreiding* ✓ answer for x / *antwoord vir x* ✓ answer for *y* / *antwoord vir y* **OR / *OF*** ✓ expanding / *uitbreiding* ✓ method / *metode* ✓ simplification / *vereenvoudiging* ✓ answer for x / *antwoord vir x* ✓ answer for *y* / *antwoord vir y*	(5)
4.5			✓ $\frac{x}{120}$ ✓ $\frac{x}{160}$ ✓ equation for time / *vergelyking vir tyd* ✓ simplification / *vereenvoudiging* ✓ simplification / *vereenvoudiging* ✓ answer / *antwoord*	(6)
				[29]

Distance/*Afstand*	Speed/*Spoed*	Time/*Tyd*
x	120 km/h	$\frac{x}{120}$
x	160 km/h	$\frac{x}{160}$

$\therefore \frac{x}{120} - \frac{x}{160} = 2hrs$
$\times 480 : 4x - 3x = 960$
$x = 960\ km$

∴ Slower truck takes/Stadiger tron neem
960 ÷ 120 = 8 hrs

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QUESTION/*VRAAG 5*				
Ques./ Vraag	**SOLUTION/*OPLOSSING***		**EXPLANATION/ *VERDUIDELIKING***	
5.1	5.1.1	Hairdryer cost / *Haardroër koste* = £100 × $\frac{R17{,}58}{£1}$ = $R1\,758$	✓ method / *metode* ✓ answer / *antwoord*	(2)
	5.1.2	Hairdryer cost / *Haardroër koste* = £100 × $\frac{R\ 15{,}00}{£1}$ = $R1\,500$	✓ answer / *antwoord*	(1)
	5.1.3	Savings = $R1\,758 - R1\,500 = R258$	✓ answer / *antwoord*	(1)
5.2	5.2.1	Deposit / *deposito* = $R10\,000 \times 12\%$ = $R1\,200$	✓ method / *metode* ✓ answer / *antwoord*	(2)
	5.2.2	$A = P(1 + in)$ $A = 8\,800(1 + 25\%(5))$ $A = R19\,800$ $\therefore Total = R19\,800 + R1\,200$ $= R21\,000$	✓ substitution into correct formula / *vervanging in korrekte formule* ✓ R8 800 ✓ answer / *antwoord* ✓ answer / *antwoord*	(4)
	5.2.3	Monthly installment / *maandelikse betaling* =$R19\,800 \div (5 \times 12)$ $= R330$	✓ method / *metode* ✓ answer / *antwoord*	(2)
5.3	$A = P(1 + i)^n$ $A = R6\,000(1 + 15\%)^5$ $A = R12\,068{,}14$		✓ correct formula / *korrekte formule* ✓ substitution into correct formula / *vervanging in korrekte formule* ✓ answer / *antwoord*	(3)
				[15]

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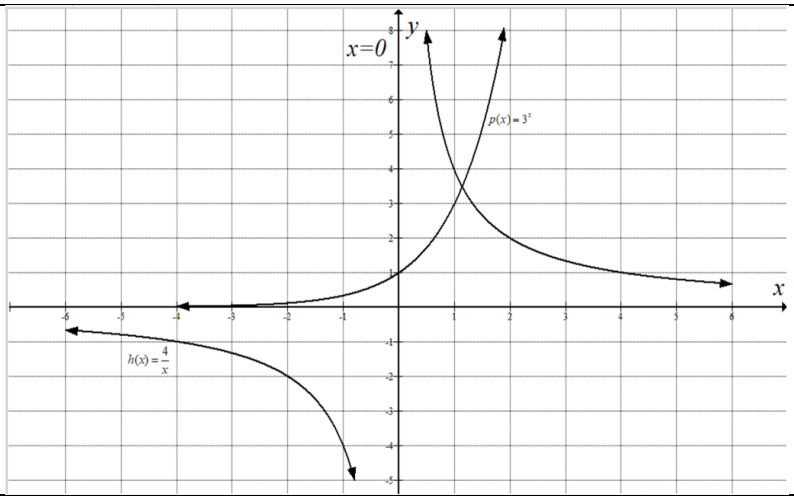
QUESTION/*VRAAG 6*

Ques./ Vraag	SOLUTION/*OPLOSSING*	EXPLANATION/ *VERDUIDELIKING*	
6.1.1			

x	-4	-3	-2	-1	0	1	2	3	4
$h(x)$	-1	$\frac{4}{-3}$	-2	-4	∞	4	2	$\frac{4}{3}$	1
$p(x)$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81

h(x):
✓ Correct values
✓ Undefined value
p(x):
✓ Correct values (3)

6.1.2		*h(x)*: ✓ shape / *vorm* ✓ quadrants / *kwadrante* ✓ asymptotes / *asymptote* *p(x):* ✓ shape / *vorm* ✓ *y*-intercept / *y-afsnit*	(5)

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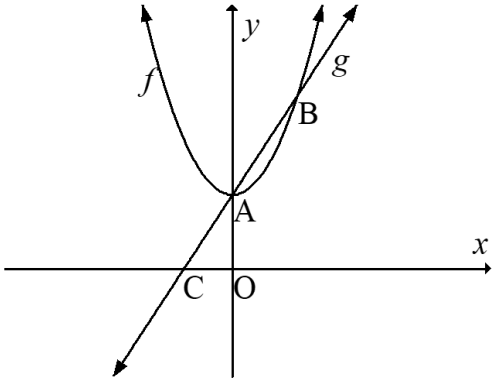
6.2			
	A: $f(x) = x^2 + 3$ $y = (0)^2 + 3$ $y = 3$ $\therefore A(0;3)$ C: $g(x) = 2x + 3$ $0 = 2x + 3$ $-3 = 2x$ $\frac{-3}{2} = x$ $\therefore C\left(\frac{-3}{2};0\right)$ B: $f(x) = g(x)$ $x^2 + 3 = 2x + 3$ $x^2 - 2x + 3 - 3 = 0$ $x^2 - 2x = 0$ $x(x-2) = 0$ $x = 0$ of $x = 2$ $\therefore x = 2$ $y = 2(2) + 3$ $y = 7$ $\therefore B\ (2;7)$	✓ $x = 0$ ✓ $A(0;3)$ ✓ $g(x) = 0$ ✓$C\left(\frac{-3}{2};0\right)$ ✓ $f(x) = g(x)$ ✓ standard form / *standaardvorm* ✓$x = 2$ ✓ $B\ (2;7)$	(8)
6.3.1	$x \in \mathbb{R}$	✓ answer / *antwoord*	(1)
6.3.2	$y \geq 3$	✓ answer / *antwoord*	(1)
6.4.1	$x > \frac{-3}{2}$	✓ Inequality / *ongelykheid* ✓ Endpoint / *Eindpunt* From coordinates of C in **6.2.1** *Vanaf koördinate van C* in **6.2.1**	(2)
6.4.2	$x = 0$ or/*of* $x = 2$	From/*Vanaf* **6.2.1** ✓ Answer / *antwoord* ✓ Answer / *antwoord*	(2)
			[22]
		TOTAL/*TOTAAL*:	**100**

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